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Roll No

EX-701-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Power System Protection

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Write the universal torque equation. 7
b) State any two advantages and disadvantages of microprocessor based relays. 7
2. a) How does a Buchholz relay work? 7
b) State drawbacks of a radial electrical distribution system. How these drawbacks are overcome in ring main electrical distribution system? 7
3. a) On what basis, the protection zones in a transmission line are categorized? 7
b) Explain any two methods by which arc can be quenched in circuit breaker. 7
4. a) Explain how a circuit breaker is rated. 7
b) State any two reasons for generation of over voltages in a power system. 7
5. a) What is factor of earthing? 7
b) Explain the construction and operation of differential relay and explain how these relays are helpful in protecting generator windings. 7

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- 6. a) Explain various types of faults that occur in an electrical generator. 7
- b) Explain in detail the protection scheme for ground fault protection in the electrical generator. 7
- 7. a) What are the functions of protective relays. 7
- b) What is back up protection? 7
- 8. a) Mention any two applications of differential relay. 7
- b) Write short notes on Circuit breaker. 7
